

IP Addressing and Subnetting

1. IP Address Classes

Class	IP Range	Example Device/Service
A	1.0.0.0 – 126.255.255.255	Large company network
B	128.0.0.0 – 191.255.255.255	University network
C	192.0.0.0 – 223.255.255.255	Home Wi-Fi router
D	224.0.0.0 – 239.255.255.255	Video streaming multicast
E	240.0.0.0 – 255.255.255.255	Research and testing

2. IP Address Calculation

Given:

- IP Address = 192.168.10.45
- Subnet Mask = 255.255.255.0 (/24)

Step 1: Network Address

Set host bits to 0.

Network Address = 192.168.10.0

Step 2: Broadcast Address

Set host bits to 1.

Broadcast Address = 192.168.10.255

Step 3: Valid Host Range

- First Host = 192.168.10.1
- Last Host = 192.168.10.254

Final Answer

Item	Address
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Host	192.168.10.1
Last Host	192.168.10.254

3. Subnetting Plan for Food Delivery Startup

Using **10.0.0.0/24**

Split into four **/26** subnets:

Group	Network Address	Subnet Mask
Delivery Partners	10.0.0.0/26	255.255.255.192
Restaurants	10.0.0.64/26	255.255.255.192
Customers	10.0.0.128/26	255.255.255.192
Admins	10.0.0.192/26	255.255.255.192

Each subnet supports **62 hosts**.

4. College Wi-Fi Subnetting

Given:

- Network = 172.16.0.0/16
- Need at least 20 subnets

Step 1: Borrow Bits

Need 20 subnets.

- $2^4 = 16$ (not enough)
- $2^5 = 32$ (enough)

Borrow **5 bits**.

Step 2: New Subnet Mask

Original = /16

New = /16 + 5 = **/21**

Subnet Mask = **255.255.248.0**

Step 3: Hosts Per Subnet

Remaining host bits = $32 - 21 = 11$

Hosts = $2^{11} - 2$

= $2048 - 2$

= **2046 hosts**

Final Answer

Item	Value
New Subnet Mask	255.255.248.0 (/21)
Number of Subnets	32
Hosts per Subnet	2046

5. Invalid IP Address Explanation

The IP address **192.168.1.300** is invalid because each number (octet) in an IPv4 address must be between **0** and **255**.

Here, **300** is greater than **255**, so the device cannot use this address.

Valid Examples in the Same Subnet

- 192.168.1.10
- 192.168.1.50
- 192.168.1.200

A valid IP in this subnet should look like:

192.168.1.x (where x is between 1 and 254).